



Database Foundations

2-3

Entities and Attributes



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Objectives

- This lesson covers the following objectives:
 - Identify entities
 - Identify attributes
 - Identify mandatory, optional, volatile, and nonvolatile attributes
 - Describe the Barker, Bachman, and Information Engineering notations



Entity

- Information that must be tracked
- Name for things that you can list (usually in noun form)



EMPLOYEE



DEPARTMENT



Entities are categories of things that are important for a business and about which information must be kept. Entities contain facts and information that the business must know and remember. Some examples of entities might include the following:

- PERSON: Agent, insured, employee, customer
- PLACE: State, country, municipality
- THING: Inventory item, vehicle, product
- CONCEPT: Policy, risk, coverage, job
- ORGANIZATION: Agency, department
- EVENT: Service request, claim, election

Identifying Entities: Example



District Hotel Manager Sharon Ferguson wants to manage the information that her company keeps about their hotels, guests, and rooms. Sharon is responsible for multiple hotels in various locations. She wants to know what rooms she has in each hotel and what the price is for each room. She also wants to know which rooms have been reserved for a particular date by a guest.



Entity Types

- An entity can be classified as one of the following types:

Name	Description	Example
Prime	Exists independently	CUSTOMER, INSTRUCTOR
Characteristic	Exists because of another (prime) entity	ORDER, CLASS OFFERING
Intersection	Exists because of two or more entities	ORDER ITEM, CLASS ENROLLMENT



Entities and Instances

- Entities contain instances
- An entity instance is a single occurrence of an entity
- Entities represent a set of instances that are of interest to a particular business



Entity	Instance
PERSON	John Smith
PRODUCT	2.5 x 35 mm copper nail
PRODUCT TYPE	Nail
JOB	Violinist

- An instance of one entity may be an entity in its own right. For example, the "violinist" instance of the JOB entity could be the name of another entity with instances like Tim Kary or Jerry Gam.

At the end of the entity modeling process, entities are typically transformed into tables, with each row of a table representing an individual instance.

Project Exercise 1

- DFo_2_3_1_Project
 - Oracle Baseball League Store Database
 - Identifying Entities



Attributes

- Attributes describe entities and are the specific information that must be known
- It is a single-valued property detail of an entity



Attributes describe an entity by qualifying, identifying, classifying, quantifying, or expressing the state of the entity. Attributes represent a type of description or detail, not an instance.

Attribute names are singular. An attribute value can be a number, a character string, a date, or an image. Every attribute has a specific data type in the database.

Attribute Characteristics, 1 of 2

- Attributes are shown within the entity box on the ERD
- Attribute names are singular and mixed case or lowercase
- In most cases, the name of the attribute should not include the entity's name, because attributes are qualified with the entity name



Attribute Characteristics, 2 of 2

- Attributes are classified as one of the following:
 - Mandatory (nulls are not allowed), indicated by *
 - Optional (nulls are allowed), indicated by a lowercase o



Volatile and Nonvolatile Attributes

- Volatile attributes are unstable attributes
 - Example: Age
- Nonvolatile attribute are stable attributes
 - Example: Birth Date

Nonvolatile
Attribute



EMPLOYEE
ID
First Name
Middle Name
Date of Birth



Project Exercise 2

- DFo_2_3_2_Project
 - Oracle Baseball League Store Database
 - Identifying Attributes



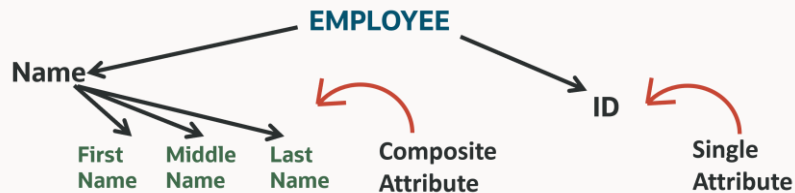
Mandatory and Optional Attributes

- Mandatory attributes must have a value
- Optional attributes may not have a value and can be blank (null)



Single and Composite Attributes

- Single or atomic attributes are attributes that cannot be divided into subparts
- Composite attributes are attributes that can be divided into smaller subparts that represent basic attributes with independent meanings of their own



DFo 2-3
Entities and Attributes

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Emp ID, age, and sex are examples of simple attributes.

Single-Valued and Multi-Valued Attributes

- Single-valued attributes can have a single value at a particular instance of time
 - Example: Student Last Name
- Multi-valued attributes can have more than one value at one time
 - Example: Address



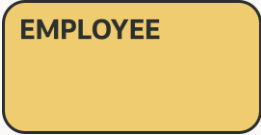
Project Exercise 3

- DFo_2_3_3_Project
 - Oracle Baseball League Store Database
 - Identifying Mandatory/Optional Attributes



Barker Notation: Rules for Drawing Entities

- An entity is represented as a round-cornered rectangle
- An entity must be named, and the name must be placed inside the entity in the upper-left corner
- The entity name should be in uppercase form
- The entity name should be in singular form



EMPLOYEE



The Barker notation is used in this course.

Barker Notation: Rules for Drawing Attributes

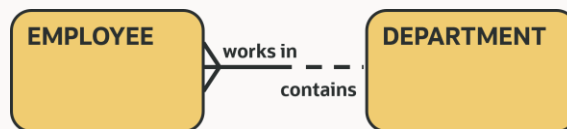
- The attributes must be written so that everyone, not just developers, can understand them
- Attributes must be written with the first letter of each word in uppercase and the rest in lowercase
- A symbol representing the type of attribute should be placed next to each attribute (* for mandatory o for optional and # for UID)

```
graph LR
    subgraph EMPLOYEE
        direction TB
        Id["# Id"]
        FN["* First Name"]
        LN["* Last Name"]
        DOB["* Date of Birth"]
        TEL["* Telephone Number"]
    end
```



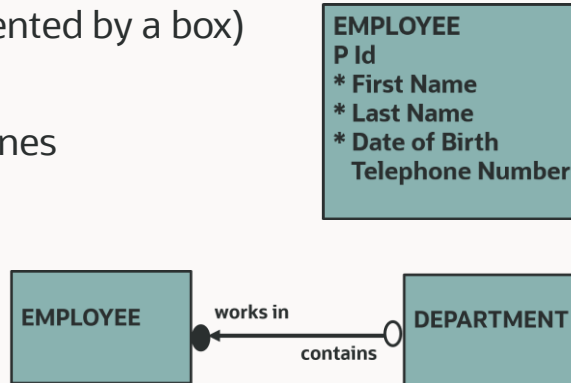
Barker Notation: Rules for Drawing Relationships

- A relationship can exist between a maximum of two entities
- A relationship can exist on the same entity (recursive)
- A relationship has two perspectives
- Both perspectives of a relationship must be labeled



Bachman Notation

- Entity (represented by a box)
- Attributes
- Relationship lines



DFo 2-3
Entities and Attributes

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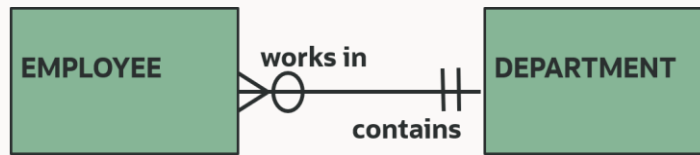
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- The Bachman notation is also a binary model.
- Attributes:
 - Show * for not null and no notation for null attributes.
 - Show P for UID and F for the attribute created through the relationship
- Relationship lines:
 - Show an arrow instead of crow's feet for the maximum cardinality
 - Show an open circle or filled-in circle instead of a dotted line and a solid line for the minimum cardinality

The relationship is depicted as a line connecting the associated entities. The relationship is given a phrase to describe the association at both ends of the line.

- In Bachman notation, a subclass is represented as an inner box within the superclass.

Information Engineering Notation

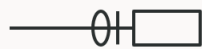
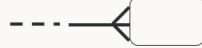
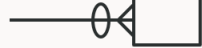


An **EMPLOYEE** works only in one **DEPARTMENT**
A **DEPARTMENT** contains zero or more **EMPLOYEES**



- The Information Engineering notation is also a binary method. It is simple and easy to read, and is well suited for high-level logical and enterprise data modeling.
- It does not allow a ternary relationship, and it does not show attributes related to a relationship. Both cardinality and participation constraints are combined into the min/max (bar and crowfoot) notation.
- The only drawback of this notation is that it does not support identification of an entity's attributes.

Data Model Notations

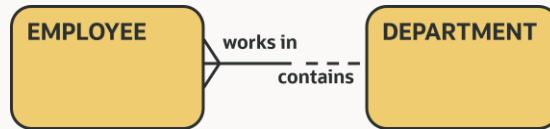
Notation (Read left to right)	Barker Notation	Bachman Notation	Information Engineering
Zero or one			
Only one			
Zero or more			
One or more			
Primary Key/Unique key	#	P	

- Note: Barker Notation is used for this course

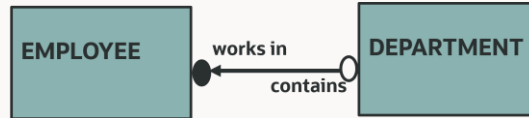


Data Model Notations: Examples

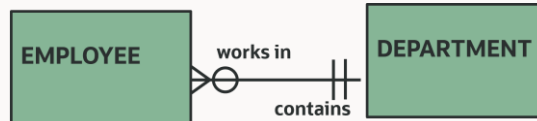
- Barker Notation



- Bachman Notation



- Information Engineering Notation



Project Exercise 4

- DFo_2_3_4_Project
 - Oracle Baseball League Store Database
 - Using Barker Notation



Summary

- In this lesson, you should have learned how to:
 - Identify entities
 - Identify attributes
 - Identify mandatory, optional, volatile, and nonvolatile attributes
 - Describe the Barker, Bachman, and Information Engineering notations



